Airline Passenger Satisfaction

Technical Pipeline Report

XGBoost-Based Dissatisfaction Risk Scoring & SHAP Insight

Test set: 25,976 passengers · Model ROC-AUC: 0.9942 · Pipeline: Dual-path preprocessing + KMeans + SHAP

Dataset: Customer Satisfaction in Airline — 129,880 rows × 22 columns

All figures generated from pipeline artifacts · No post-hoc adjustments

1. Executive Summary

- Dataset: 129,880 survey responses, 22 features, binary target (satisfied / dissatisfied), class balance 54.7% / 45.3% — no significant imbalance; SMOTE was not applied.
- Production model: XGB-Base (XGBoost + Isotonic calibration, Optuna 20-trial tuning). Test-set ROC-AUC = **0.9942**, F1 = 0.9625, Accuracy = 0.9593 at threshold 0.459 (Youden's J).
- Risk distribution: **40.0%** of the test set (10,390 passengers) fall in the High or Critical risk tier. Both tiers each account for exactly 20% by design (quintile split).
- Primary dissatisfaction drivers (SHAP): **Seat comfort** (rank 1 by SHAP, rank 7 by bivariate correlation — non-linear effect) and **Inflight entertainment** (rank 2/3). Departure Delay accounts for only 1.5% of Critical-group top-3 drivers.
- Loyalty bias finding: Within cluster C1 (Tr■i Nghi■m Kém), disloyal customers have mean risk_score 0.811 vs 0.627 for loyal customers, yet service ratings differ by <0.1 points across all dimensions — the score differential is attributable to the Customer Type feature itself, not measured service quality.

2. Dataset & Preprocessing

2.1 Data Source

Source: Customer Satisfaction in Airline survey export (2026-06-16). 129,880 rows × 22 columns. One row per flight experience. Target variable: *satisfaction* — binary, no intermediate scale. No customer ID; rows treated as independent observations.

2.2 Preprocessing Decisions

The following table summarises all pre-processing decisions taken in Phase 1 (EDA) and Phase 2 (feature engineering) of the pipeline, together with the technical justification for each.

Decision	Detail	Justification
Drop Arrival Delay	Remove Arrival Delay in Minutes	Pearson r = 0.9653 with Departure Delay → multicollinearity; keeping both adds no information and inflates linear mode
Recode rating 0 → NaN	14 Likert rating cols (1–5 scale): 0 recoded to NaN	Verified N/A hypothesis: satisfaction rate at rating=0 is 0.447, close to rating=3 (0.510) and far from rating=1 (0.268), indicating 0 =
Create was_na_* indicators	14 binary cols: was_na_<feat>=1 if original=0	Preserves information that the service was not used/applicable — a structurally different signal from low satisfaction
Impute ratings — Tree path	14 rating cols: NaN kept as-is for XGBoost	XGBoost handles missing values natively via learned split direction; imputing would erase the N/A signal.
Impute ratings — Linear path	14 rating cols: median imputation (fit on train)	LogisticRegression cannot handle NaN; median chosen over mean to be robust to the skewed rating distributio
Composite score imputation	3 composite cols: median imputation (fit on train)	KMeans (sklearn) requires non-NaN input; composite scores had 0% NaN (mean of 5–7 cols), so imputation is preca
Clip Departure Delay (linear path only)	43 rows (0.03%) have delay > 500 min — cap at p99.9 of train = 181 min	Extreme outliers distort MinMaxScaler range. XGBoost handles these natively via tree splits, so clip
Ordinal encoding — Class	Eco=0, Eco Plus=1, Business=2	Natural ordinal relationship preserved. OHE would create spurious dummy trap and increase dimensionality without adding
Binary encoding — Customer Type	Loyal=1, Disloyal=0	Binary categorical; no ordinal relationship assumed between 0/1.
Binary encoding — Type of Travel	Business travel=1, Personal=0	Binary categorical.
Train/Test split	Stratified 80/20, random_state=42	Stratification ensures class balance preserved in both sets (54.7% satisfied in both train and test).

3. Feature Engineering

3.1 Composite Experience Scores

Three composite features were engineered to provide aggregate experience signals. These replace the RFM (Recency–Frequency–Monetary) segmentation used in the E-Commerce pipeline, which is inapplicable here: the dataset contains one survey response per flight with no temporal transaction history, making Recency and Frequency undefined and Survival Analysis without a time-to-event endpoint.

Feature	Formula	Component Columns (n)	NaN in Train
ground_experience_score	mean(GROUND_COLS, skipna=True)	Ease of Online booking, Online boarding, Checkin service, Gate location, Baggage handling (5)	0 (0.0%)
inflight_experience_score	mean(INFLIGHT_COLS, skipna=True)	Seat comfort, Food and drink, Inflight wifi service, Inflight entertainment, On-board service, Leg room service, Cleanliness (7)	0 (0.0%)
delay_severity	Departure Delay / (Flight Distance + 1)	Departure Delay in Minutes, Flight Distance (2)	0 (0.0%)

Zero NaN in all three composite features: no customer had all columns in a group simultaneously missing, so mean(skipna=True) always returned a value.

4. Dual Preprocessing Pipeline

Two parallel preprocessing paths were maintained to support fundamentally different model families — tree-based (XGBoost, Random Forest, Decision Tree) and linear (Logistic Regression). Mixing these paths would either impair tree models (unnecessary imputation destroys the N/A signal) or break linear models (NaN inputs not supported).

Step	Tree Path (XGB / RF / DT)	Linear Path (LR)
Outlier handling	None (tree splits handle extremes)	Departure Delay clipped at p99 train = 181 min
Rating imputation	NaN kept — XGBoost uses learned split direction	14 rating cols → median (fit on train)
Composite imputation	Median (fit on train) — required for KMeans	Median (fit on train)
Encoding	Ordinal/binary (as Section 2.2)	Same
Scaling	None needed for tree models	MinMaxScaler [0,1] on all numeric cols (was_na_* excluded)
Feature selection	All features passed to model	Chi² SelectKBest, $p < 0.05$ → 30/38 kept (Base), 32/40 (Dist)
KMeans features	cluster_id or centroid distances added as features	Centroid distances (dist_centroid_0/1) added in Dist variant
SMOTE	Skipped — minority class 45.3% ≥ 40% threshold	Skipped (same reason)

KMeans Feature Engineering

KMeans (K=2, selected by silhouette score among K=2..8) was fit exclusively on the training set using 5 features: ground_experience_score, inflight_experience_score, delay_severity, Age, Flight Distance. Categorical features were excluded to ensure clusters reflect service experience rather than demographic segments.

Two strategies were produced: Strategy A (Euclidean distances to each centroid — 2 columns) for the linear path, and Strategy B (cluster_id integer) for the tree path. Test set was transformed via predict() on the fitted model — no refit.

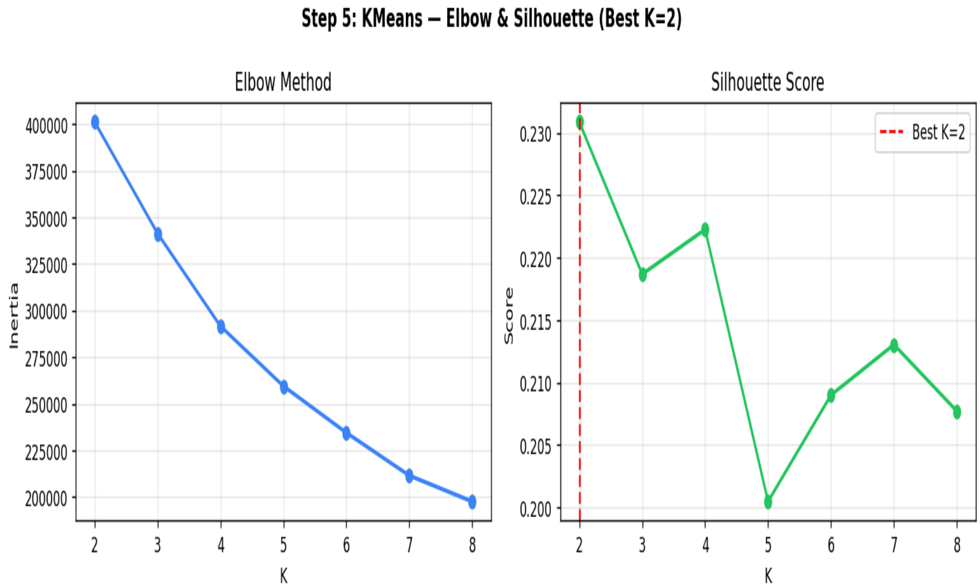

Figure 4.1: Elbow and Silhouette plots for KMeans K=2..8. Silhouette peaks at K=2 (0.2309), confirming 2 clusters.

5. Model Comparison

Model	Threshold	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
XGB-Base	0.459	0.9593	0.9712	0.9539	0.9625	0.9942	0.9955
XGB-Clust	0.445	0.9588	0.9679	0.9565	0.9622	0.9943	0.9956
RF-Clust	0.499	0.9468	0.9584	0.9437	0.951	0.9905	0.9929
DT-Clust	0.548	0.9249	0.9415	0.9199	0.9306	0.9809	0.9844
LR-Base	0.558	0.8647	0.887	0.8626	0.8747	0.9414	0.9526
LR-Dist	0.541	0.8649	0.8826	0.8687	0.8756	0.9414	0.9526

Table 5.1: Full 6-model comparison on test set (n=25,976). Threshold optimised via Youden's J on training predictions. XGB-Base and XGB-Clust calibrated via CalibratedClassifierCV (isotonic, cv=5).

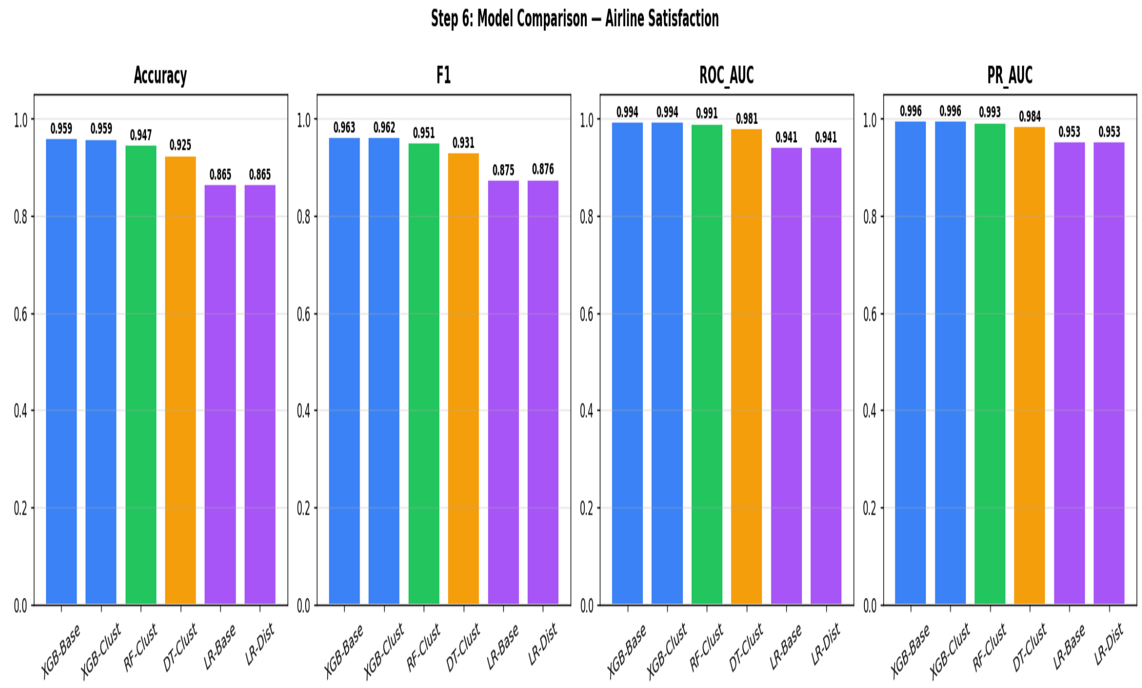

Figure 5.1: Accuracy, F1, ROC-AUC, and PR-AUC for all 6 models.

Production Model Selection: XGB-Base

XGB-Base was selected as the production model based on the following observations:

- XGB-Clust vs XGB-Base: ROC-AUC 0.9943 vs 0.9942 — improvement of +0.0001, negligible. Adding KMeans cluster_id does not improve XGBoost's predictive ability, likely because XGBoost can learn the cluster structure implicitly through its own splits.
- LR-Dist vs LR-Base: ROC-AUC 0.9414 both — KMeans distance features provide zero lift to Logistic Regression either.
- KMeans clusters are retained for profiling and risk scoring (Sections 6–8), but not incorporated into the production prediction model.
- XGB-Base was tuned with Optuna TPE sampler (20 trials, 5-fold stratified CV). Best params: n_estimators=203, max_depth=8, lr=0.127, subsample=0.947, colsample_bytree=0.592. CV ROC-AUC = 0.9938.

6. Cluster Profiling — Passenger Personas

uster	Persona	N	Dissatisfied %	Ground Score	Inflight Score	Delay Severity	Age (mean)	Biz Travel %	Loyal
	Hài Lòng Toàn Di■n	14,135	22.8%	3.871	3.820	0.0110	41.6	68.5%	87.0%
	Tr■i Nghi■m Kém	11,841	72.1%	2.776	2.757	0.0110	37.1	69.0%	75.3%

Table 6.1: Cluster profile summary from KMeans K=2 (test set, n=25,976).

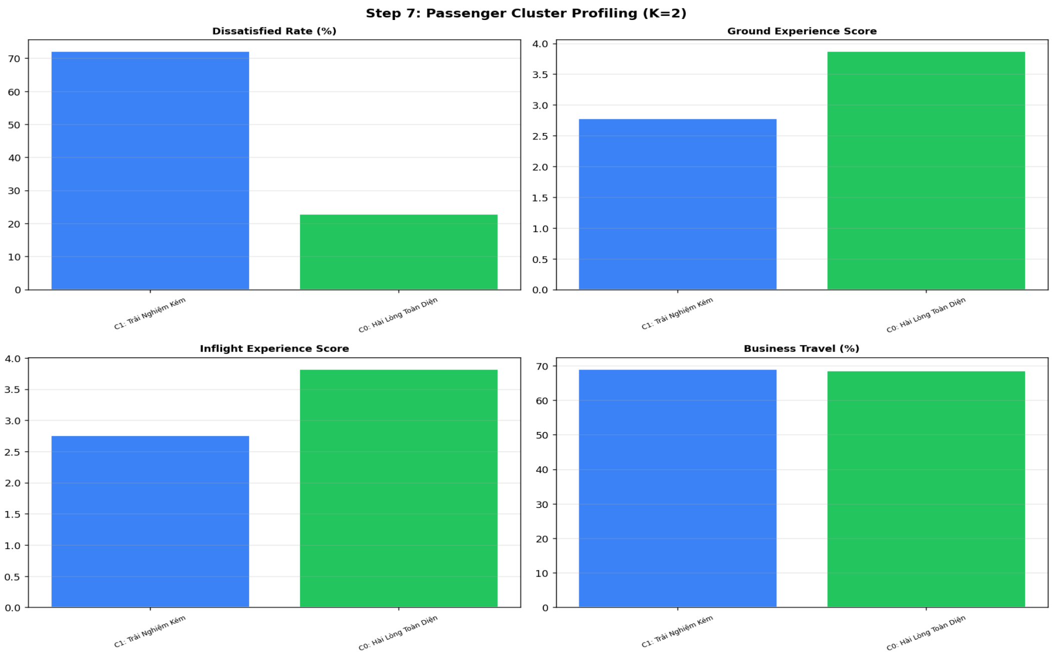

Figure 6.1: Cluster comparison — Dissatisfied rate, Ground score, Inflight score, and Business Travel %.

Key Finding: Personas Reflect Service Quality, Not Demographics

Both clusters have nearly identical Business Travel proportions (C0: 68.5%, C1: 69.0%) and similar delay severity (0.011 for both). The primary differentiator is service experience: C1 has ground_score 2.78 vs 3.87 for C0 (Δ = –1.09) and inflight_score 2.76 vs 3.82 (Δ = –1.06). The cluster structure captures satisfaction level, not passenger segment.

7. Risk Scoring

7.1 Formula

```
passenger_value_proxy = (Class_ord / 2 + biz_travel) / 2
# Class_ord: Eco=0, Eco Plus=1, Business=2 (÷2 → [0,1])
# biz_travel: 1 if Business travel, 0 if Personal
```

```
RawScore = 0.7 × P(dissatisfied) + 0.3 × passenger_value_proxy
RiskScore = (RawScore − min) / (max − min) # min-max by test set
```

The 0.7/0.3 weights assign primary weight to the model's dissatisfaction probability and secondary weight to passenger commercial value (business-class / business-travel passengers are weighted higher as proxies for higher lifetime value, despite the dataset lacking actual spend data).

7.2 Risk Tier Distribution

Tier	Threshold Range	Count	% of Test Set
Very Low	0.000 – 0.3000	8,538	20%
Low	0.3000 – 0.3000	3,223	~12%
Medium	0.3000 – 0.6775	3,825	~15%
High	0.6775 – 0.8496	5,199	20%
Critical	0.8496 – 1.000	5,191	20%
High + Critical	—	10,390	40%

■ *Technical note: quintile-based split means each tier's percentage is fixed by design at ~20%, not derived from the data distribution. The informative values are the threshold cutpoints (p20=0.300, p40=0.300, p60=0.678, p80=0.850), not the % counts. The near-identical p20 and p40 thresholds indicate a high density of low-risk scores — the distribution is right-skewed.*

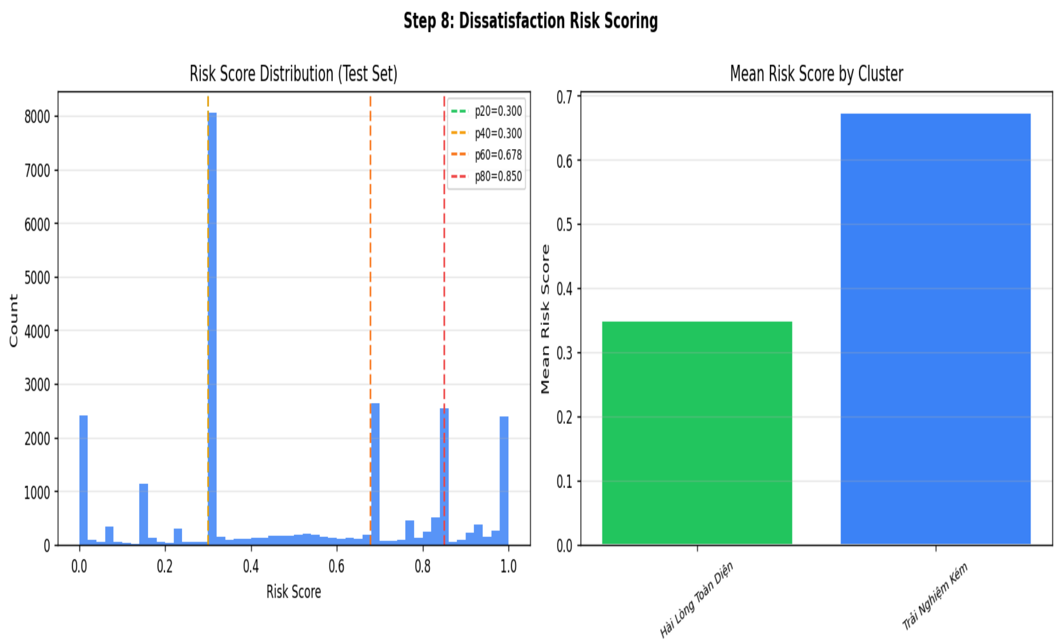

Figure 7.1: Risk score distribution (left) and mean risk score by cluster (right).

8. SHAP Analysis

8.1 Global Feature Importance

SHAP TreeExplainer was applied to the base XGBoost estimator extracted from CalibratedClassifierCV. SHAP values were computed for all 25,976 test rows. Feature importance = mean(|SHAP value|) across all rows.

SHAP Rank	Feature	Mean |SHAP|	PtBis Rank	Δ Rank
1	Seat comfort	1.7207	7	-6 ■
2	Inflight entertainment	0.9934	1	+1
3	loyal	0.7138	—	—
4	biz_travel	0.5217	—	—
5	Departure/Arrival time convenient	0.5175	—	—
6	inflight_experience_score	0.4527	2	+4 ■
7	Ease of Online booking	0.3783	4	+3 ■
8	Gate location	0.3573	—	—
9	Baggage handling	0.3131	—	—
10	Online support	0.2847	5	+5 ■

Table 8.1: SHAP global importance vs Point-Biserial rank (Step 3). ■ = rank shift ≥ 3 positions, indicating non-linear effect.

Notable rank shift: Seat comfort moves from PtBis rank 7 to SHAP rank 1 — the strongest feature in the model. This indicates a non-linear interaction effect that bivariate correlation does not capture. Similarly, *loyal* (Customer Type) and *biz_travel* do not appear in Point-Biserial top-10 (as binary indicators they were not included in that analysis) but rank 2nd and 4th by SHAP globally.

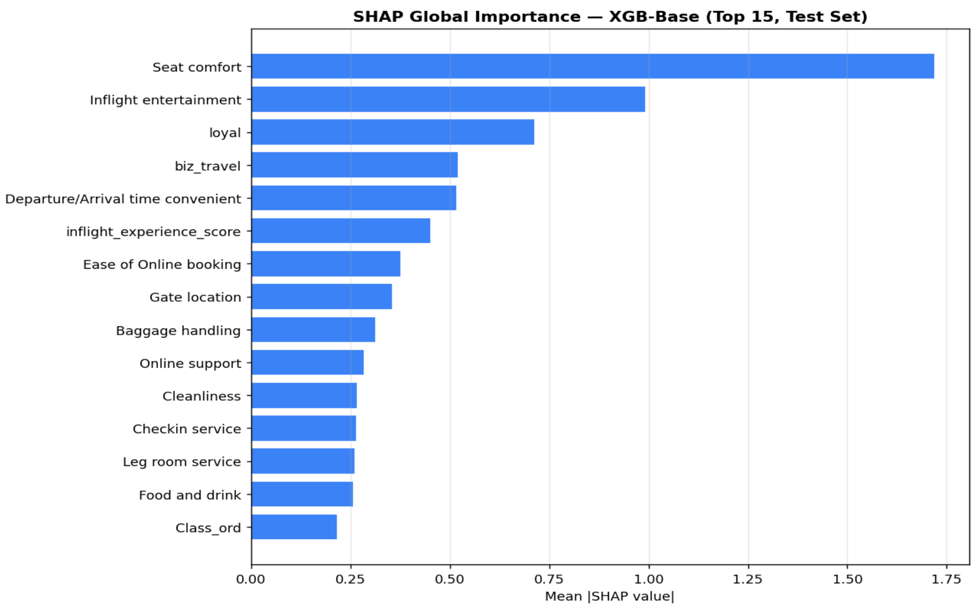

Figure 8.1: SHAP global importance bar chart — Top 15 features, test set.

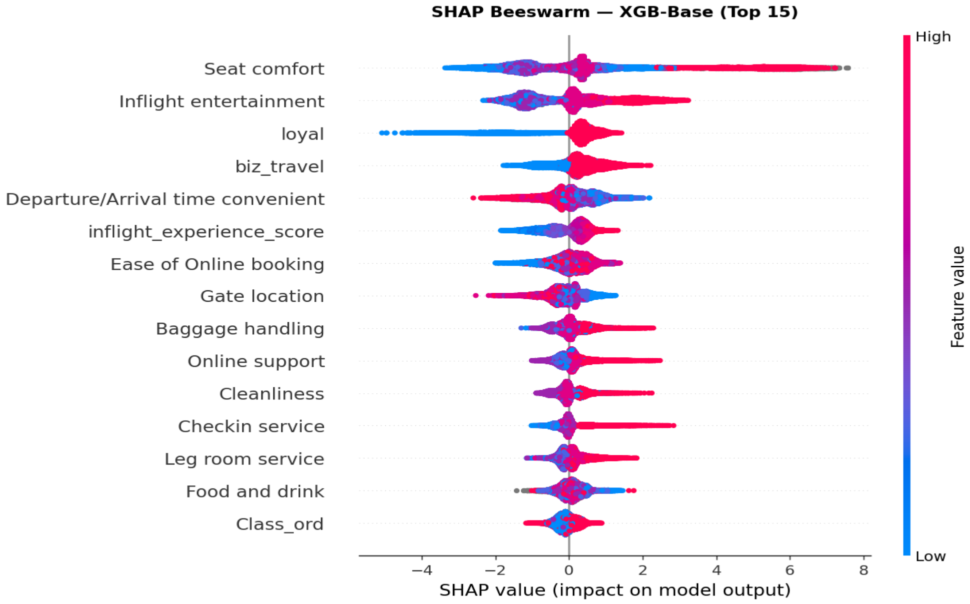

Figure 8.2: SHAP beeswarm — direction and magnitude of feature effects. Red = high feature value, Blue = low. Positive SHAP = pushes toward satisfied.

8.2 SHAP by Cluster

SHAP values were computed separately for C0 and C1 sub-populations to identify whether the primary drivers differ between satisfied and dissatisfied clusters.

Rank	C0 Feature	C0 Mean|SHAP|	C1 Feature	C1 Mean|SHAP|
1	Seat comfort	1.8397	Seat comfort	1.5785
2	Inflight entertainment	1.0130	Inflight entertainment	0.9699
3	biz_travel	0.6532	loyal ←	0.7903
4	loyal	0.6497	inflight_experience_score ←	0.5426
5	Departure/Arrival time convenient	0.5224	Departure/Arrival time convenient	0.5117

Table 8.2: Top-5 SHAP features per cluster. ← marks rank-differing positions.

Key difference: In C0 (satisfied cluster), *biz_travel* ranks 3rd — the travel purpose modulates satisfaction for passengers who otherwise rate services well. In C1 (dissatisfied cluster), *loyal* rises to 3rd, reflecting that Customer Type has strong predictive weight within the group with uniformly low service ratings.

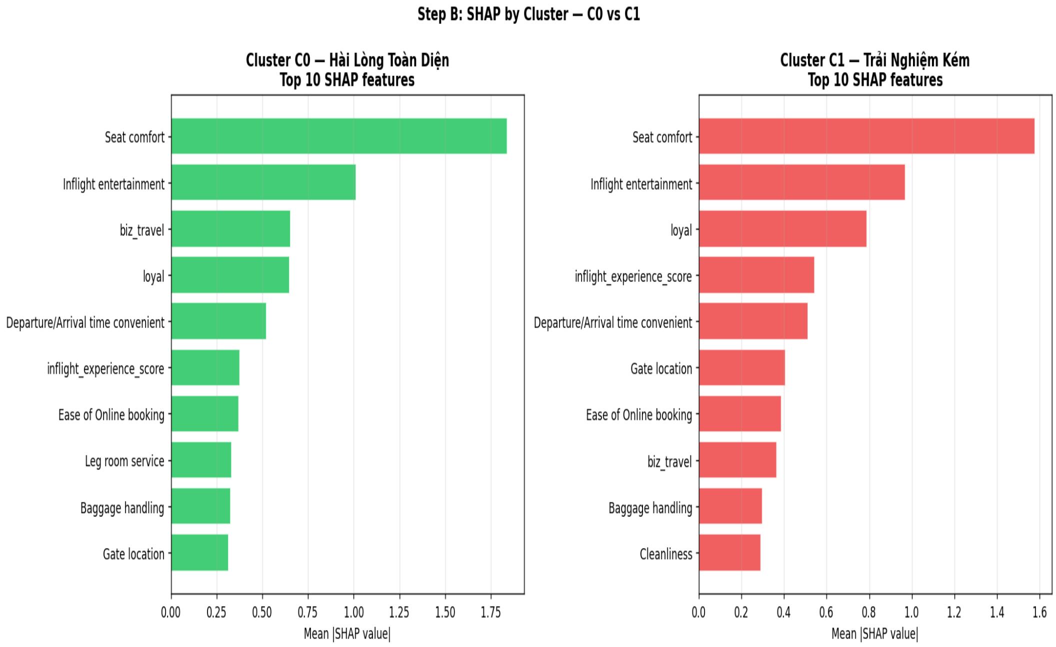

Figure 8.3: SHAP top-10 by cluster — C0 (green) vs C1 (red).

8.3 Critical Group SHAP

SHAP analysis restricted to the Critical risk tier (n=5,191, 20% of test set).

Driver Feature	% Critical	Mean Rating (Critical)	Mean Rating (All)	Δ
Seat comfort	86.3%	2.16	2.95	-0.79
Inflight entertainment	68.2%	2.43	3.46	-1.03
loyal	48.6%	0.51	0.82	-0.31
inflight_experience_score	26.7%	2.69	3.34	-0.65
Gate location	9.8%	3.07	2.99	+0.08
Ease of Online booking	9.6%	2.68	3.47	-0.79
Departure/Arrival time convenient	9.0%	—	—	—
Age	7.8%	—	—	—

Table 8.3: Driver frequency and mean actual ratings in Critical group vs full test set.

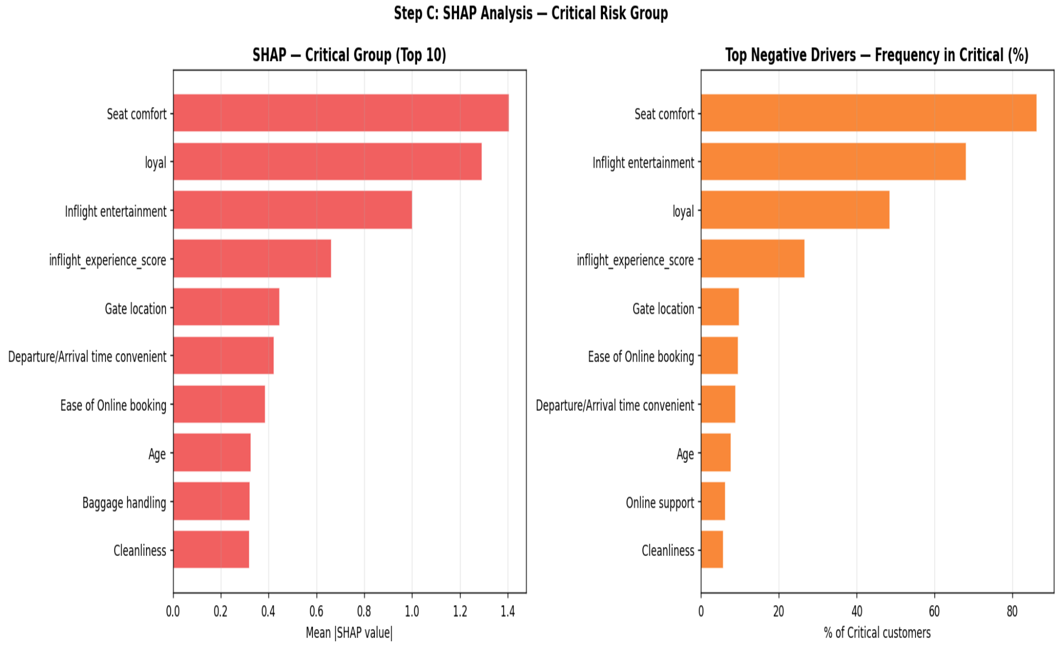

Figure 8.4: SHAP top-10 for Critical group (left) and driver frequency in Critical top-3 (right).

9. Key Finding: Loyalty Bias in Risk Score

Within cluster C1 (Tr■i Nghi■m Kém, n=11,841), passengers are split by Customer Type: 75.3% Loyal (n=8,916) and 24.7% Disloyal (n=2,925). The risk score distribution between these sub-groups shows a statistically meaningful difference that warrants methodological scrutiny.

Metric	Loyal (C1)	Disloyal (C1)	Difference
Count	8,916 (75.3%)	2,925 (24.7%)	—
Mean risk_score	0.627	0.811	–0.184
High/Critical rate	61.1%	86.2%	–25.1 pp
Seat comfort (mean rating)	2.37	2.40	+0.03
Inflight entertainment	2.93	2.42	–0.51
inflight_experience_score	2.76	2.74	–0.02
ground_experience_score	2.75	2.86	+0.11
On-board service	2.76	2.95	+0.19
Leg room service	2.89	2.99	+0.10

Table 9.1: C1 sub-group analysis — Loyal vs Disloyal passengers. Risk scores and High/Critical rates are model outputs; rating values are actual survey responses.

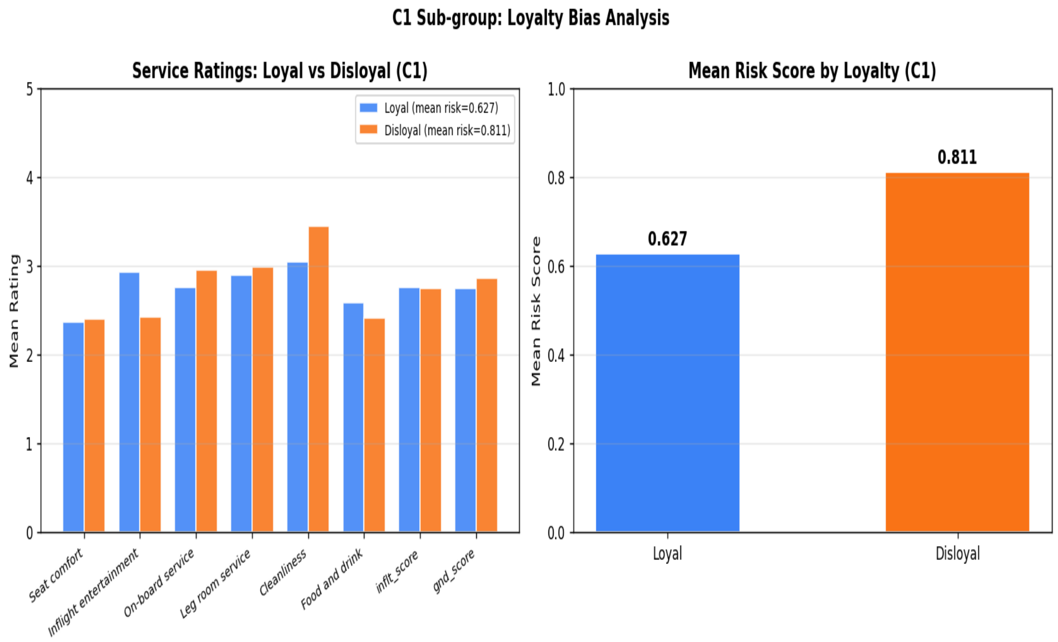

Figure 9.1: Service ratings (left) and mean risk score (right) for Loyal vs Disloyal within C1. Rating differences are minimal (<0.5 across all dimensions) while risk score differs by 0.184.

Methodological Interpretation

The risk score gap (0.627 vs 0.811) between Loyal and Disloyal passengers in C1 arises from the high SHAP weight assigned to the *loyal* feature (SHAP rank 2 globally, rank 3 within C1). However, the measured service experience — as captured by all rating columns — differs by less than 0.5 points across every dimension and is within 0.11 for composite scores.

This pattern is consistent with one or more of the following interpretations: (a) loyal customers genuinely differ in unmeasured factors (e.g., expectation calibration, repeat-flight baseline) that correlate with satisfaction but are not captured in the survey; (b) the *loyal* label carries historical satisfaction signal from past flights not present in this survey; or (c) the model has learned a spurious correlation between Customer Type and dissatisfaction outcome that does not reflect a causal service quality gap.

Without a controlled experiment isolating Customer Type from actual service experience, these interpretations cannot be distinguished from the data alone. Risk scores for Disloyal passengers in C1 should be interpreted with this uncertainty in mind.

10. Limitations & Caveats

- *RFM segmentation and Kaplan-Meier survival analysis were not applied: the dataset contains one row per survey, with no repeated transaction history and no time-to-event endpoint. Applying temporal methods to this data would constitute a methodological misuse.*
- *Risk tiers (Very Low / Low / Medium / High / Critical) are defined by quintile split on the test set population. Each tier covers exactly ~20% by construction. The tier labels imply relative risk within this population, not absolute risk benchmarks. Threshold values (p20=0.300, p40=0.300, p60=0.678, p80=0.850) are the operationally informative quantities.*
- *The loyalty bias described in Section 9 has not been isolated through controlled ablation: no model variant excluding Customer Type was trained for comparison. The magnitude of the bias attributable to the feature vs genuine behavioural differences remains unknown.*
- *SHAP values reflect marginal feature contributions within the XGBoost model (log-odds scale). They describe the model's learned associations, not causal effects. A high SHAP value for Seat comfort indicates the model relies heavily on that feature — it does not establish that improving seat comfort will causally change satisfaction outcomes.*
- *passenger_value_proxy (used in risk scoring) is a structural approximation based on ticket class and travel type. No actual spend, revenue, or lifetime value data was available. The proxy may misrank passengers whose commercial value diverges from these structural indicators.*
- *Model performance metrics are measured on a single hold-out test split (random_state=42). No repeated cross-validation or bootstrap confidence intervals were computed for the final test metrics.*

11. Appendix — Artifact File Index

Category	Filename	Description
Model	xgb_base_model.pkl	XGB-Base — CalibratedClassifierCV, production model
Model	xgb_clust_model.pkl	XGB-Clust — with KMeans cluster_id feature
Model	rf_clust_model.pkl	RF-Clust — Random Forest
Model	dt_clust_model.pkl	DT-Clust — Decision Tree
Model	lr_base_model.pkl	LR-Base — Logistic Regression (Chi² Base features)
Model	lr_dist_model.pkl	LR-Dist — Logistic Regression (Chi² Dist features)
Model	kmeans_model.pkl	KMeans K=2 (fit on train, 5 features)
Model	kmeans_scaler.pkl	StandardScaler for KMeans input features
Model	imputer.pkl	SimpleImputer (median) for 14 rating cols — linear path
Model	imputer_composite.pkl	SimpleImputer (median) for 3 composite scores
Model	scaler.pkl	MinMaxScaler — linear path
Model	chi2_meta.pkl	Chi² feature selection metadata (Base + Dist)
Model	risk_score_meta.pkl	RiskScore min/max and quintile thresholds
Data	model_comparison.csv	6-model metrics table
Data	cluster_summary.csv	KMeans K=2 cluster profile
Data	customer_risk_insight_report.csv	Per-customer risk + top-3 SHAP drivers + actual ratings
Data	critical_driver_frequency.csv	Driver frequency in Critical group
Data	shap_global_importance.csv	Global SHAP mean|value| for all 37 features
Data	system_level_insight_report.md	System-level descriptive insight report
Chart	p2_03_pearson_heatmap.png	Pearson heatmap — 14 rating cols
Chart	p2_03_point_biserial.png	Point-Biserial top-12 vs satisfaction
Chart	p2_05_elbow_silhouette.png	KMeans elbow + silhouette K=2..8
Chart	p2_05b_chi2_scores.png	Chi² feature selection — Base variant
Chart	p2_06_model_comparison.png	6-model comparison bar chart
Chart	p2_07_cluster_profiling.png	Cluster profiling — 4-metric bar chart
Chart	p2_08_risk_score.png	Risk score distribution + mean by cluster
Chart	p2_shap_bar.png	SHAP global importance bar — top 15
Chart	p2_shap_beeswarm.png	SHAP beeswarm — direction + magnitude
Chart	p2_shap_by_cluster.png	SHAP top-10 by cluster (C0 / C1)
Chart	p2_shap_critical.png	SHAP for Critical group + driver frequency
Chart	p2_loyalty_bias_c1.png	C1 loyalty bias: ratings vs risk score

All model artifacts are located under models/AIRLINE/. All charts are under outputs/AIRLINE/. All data CSVs and reports are under data/AIRLINE/.